Search and Screening Report

Remote sensing and geospatial approaches for multi-hazard monitoring in Nepal

Systematic / hybrid review - Scopus + Web of Science (2016-2025). Full manual screening.

Methods (paste-ready)

Literature was identified from Scopus and the Web of Science Core Collection using five complementary Boolean streams (one multi-hazard query and four hazard-specific queries), each combining remote-sensing/geospatial terms with hazard terms and “Nepal”, restricted to 2016-2025. The two searches returned 1,362 records (567 Scopus; 795 Web of Science). Records were pooled and de-duplicated with a reproducible logged routine using ordered keys (DOI, then title+year, then native identifier), removing 855 duplicates and leaving 507 unique records; 248 were indexed in both databases. After removing 24 non-article document types, 483 articles entered screening.

All 483 articles underwent manual title/abstract screening against a three-part eligibility test (Nepal study area; concrete remote-sensing/geospatial method; hazard-monitoring focus); a documented relevance classifier provided only an initial sort and provisional tags. Screening excluded 228 articles (160 not a Nepal study area - most centred on other High Mountain Asia regions or generic methods not applied to Nepal; 52 off-topic; 16 remote sensing not central), leaving 255 confirmed in scope. One further article, identified through citation snowballing, was added on the same basis, for 256 articles in scope. From these, a 100-article core was selected by a weighted, hazard-stratified priority score for in-depth synthesis. The full procedure is scripted and reproducible.

PRISMA record flow

Stage	n	Cumulative
Records identified - Scopus	567	
Records identified - Web of Science	795	
Total identified	1,362	1,362
Duplicates removed (DOI 843; title+year 12)	-855	507
Non-article doc types removed	-24	483
Excluded at title/abstract screening (manual)	-228	255
Added via citation snowballing	+1	256
Records carried to full-text screening	256	256
In-depth synthesis core	100	100

Stage-one screening outcome

Stage-one decision	Articles
Include	255
Added via citation snowballing	1
Exclude	228

Exclusion reason	Articles
Not a Nepal study area (other region / generic method)	160
Off-topic (air quality, ecology, social, agronomy, governance)	52

Exclusion reason	Articles
Remote sensing not central	16

Primary hazard (in-scope, n=256)	Articles
Landslide	113
Multi-hazard / cascade	69
GLOF / glacier	34
Flood	33
Drought	7

Scope of the synthesis: a two-tier design

- Quantitative layer - all 256 in-scope articles: publication trends, database coverage, hazard and sensor/method distributions.
- In-depth synthesis core - 100 articles read closely and placed in the hazard x capability maturity matrix; reached by priority ranking with a hazard floor so all five hazards are represented (Landslide 39, Multi-hazard 23, GLOF 16, Flood 15, Drought 7).

Drought contributes only 7 because that is the full set of genuine Nepal drought remote-sensing studies identified - itself a notable evidence gap.